

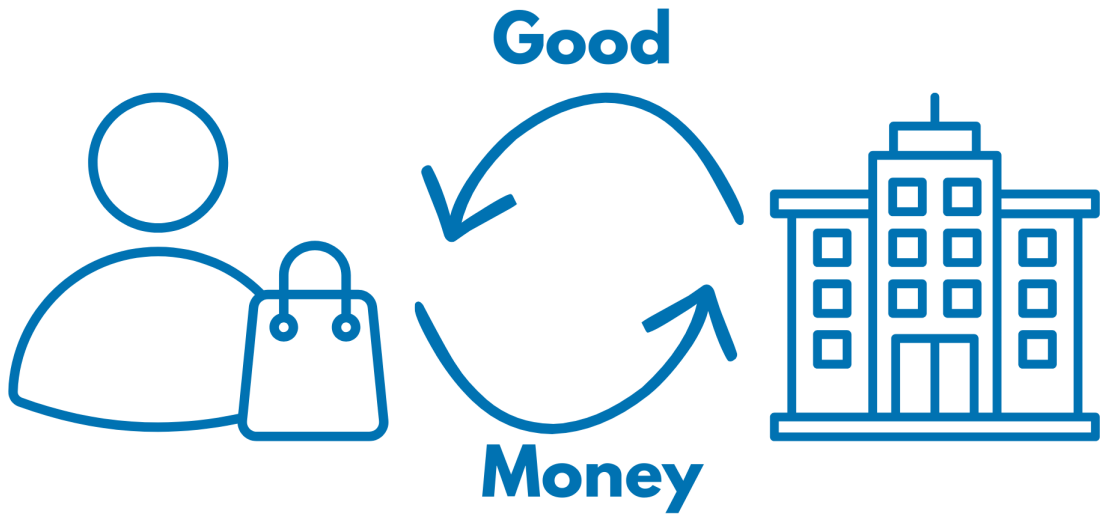
Generative AI, Online Information Quality & Supply

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Joan Lluís, Florian Oswald
Amedeo Piolatto, Andrea Gallice



Information



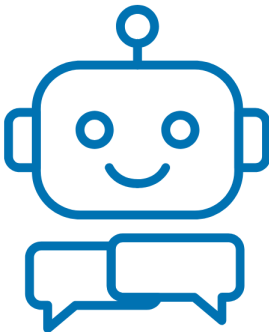
Attention



Information

←→

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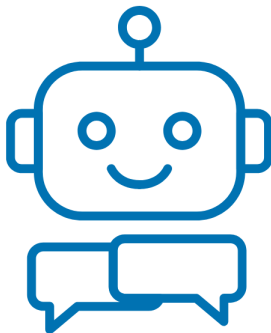




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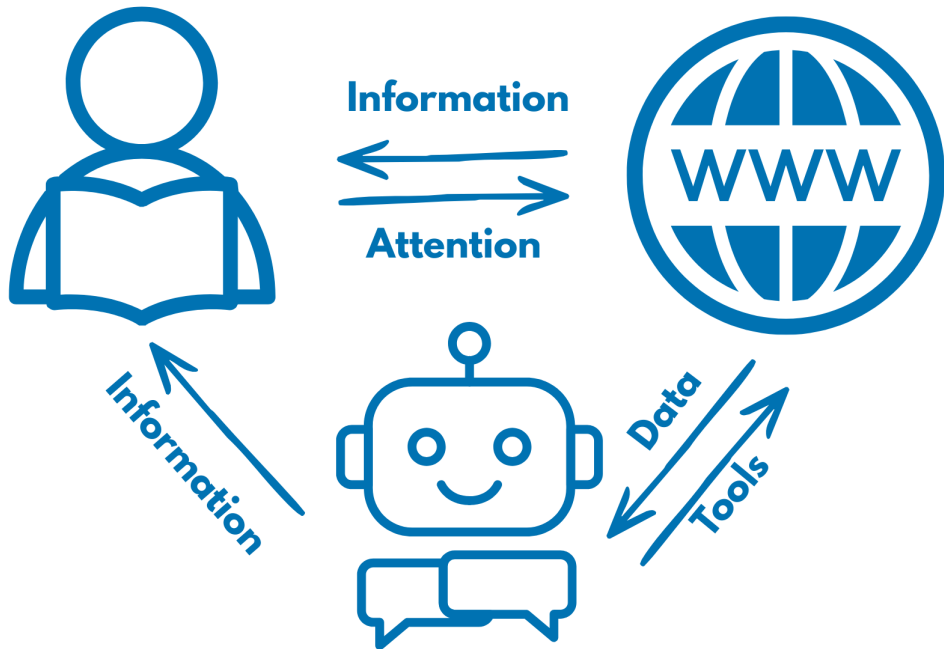
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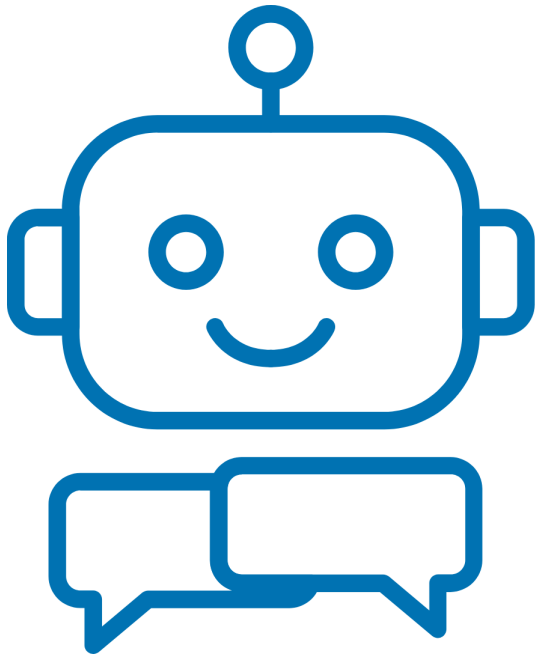


Data

↗↘

Tools





Effect 1: The AI steals customers, thus **less** incentive to produce **content**.

Effect 2: The AI makes it easier to produce, thus **more content**.

- ▶ RQ: How does **Generative AI** (Gen AI) affect **online content supply and quality**?

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 - ▶ Today: A Simple Model of the Internet:
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- ▶ **Preview:**
 - ▶ With high efficiency of AI, the Supply and the Quality increase.

Why is this question important?

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- ▶ If quality rises → better productivity, innovation, and welfare.

- ▶ **Gen AI's Impact Online**

Burtch, Lee, and Chen 2024; Rio-Chanona, Laurentsyevea, and Wachs 2023; Reeves, Yin, and Simperl 2025; Shan and Qiu 2025; Lyu et al. 2025; Zhao and Berman 2025;

- ▶ **Gen AI's Impact on Information Acquisition**

Acemoglu, Kong, and Ozdaglar 2026; Ide 2025; Agarwal, Moehring, and Wolitzky 2025; **Koren et al. 2026**; Ide and Talamàs 2025

- ▶ **Digital Technologies' Impact on Collective Information**

Dasaratha and He 2022; Acemoglu, Ozdaglar, and Siderius 2024; Agarwal, Moehring, Rajpurkar, et al. 2023; Cullen, D. Li, and S. Li 2025

- ▶ **Contribution:** (Ideally) Empirical proof of Gen AI's impact on online information stock and quality, with clear mechanisms.

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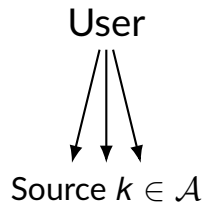
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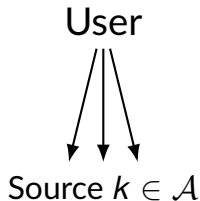
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Timing of the model:

1. Creators draw quality of their information.
2. Given the quality decide whether to enter the market or not.
3. Users pick the sources among the creators that entered.

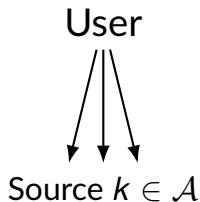




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- The shares of each source:

$$s_k = \frac{Q_k^{1/\theta}}{\sum_{j \in \mathcal{A}} Q_j^{1/\theta}}$$

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$$m_A = \frac{r(\gamma - 1/\theta)}{F\gamma}.$$

- ▶ We care about the quality that is consumed by the users:

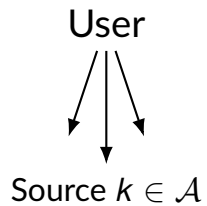
$$\tilde{Q} \equiv \sum_{j \in A} s_j Q_j$$

- We care about the quality that is consumed by the users:

$$\begin{aligned}\tilde{Q} &\equiv \sum_{j \in A} s_j Q_j \\ &= \frac{\gamma - 1/\theta}{\gamma - 1/\theta - 1} Q_0\end{aligned}$$

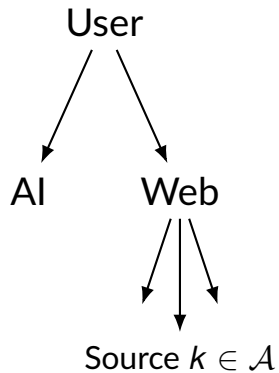
The Role of AI

1. Stealing Bussiness



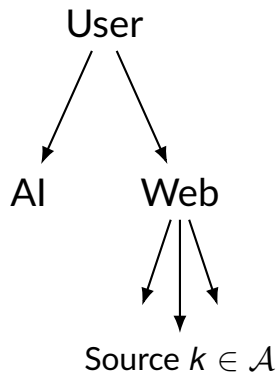
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2. Knowledge Aggregation

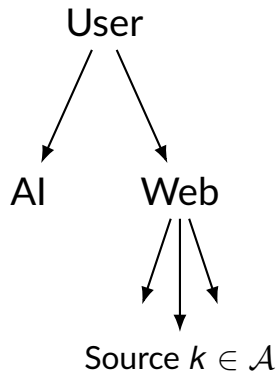
- ▶ AI takes available sources as input.
- ▶ It produces its own quality according to a CES function

$$Q_A = \phi \left(\sum_{k \in \mathcal{A}} Q_k^{1/\theta} \right)^\theta,$$

ϕ being the efficiency of AI.

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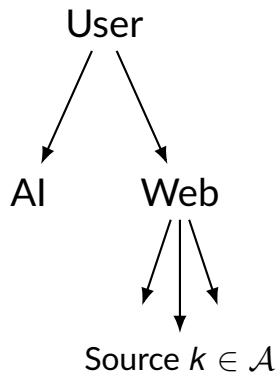
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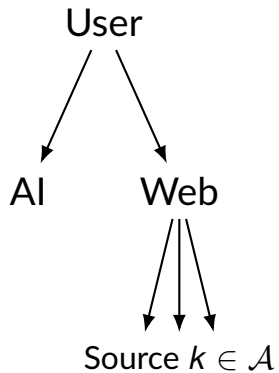
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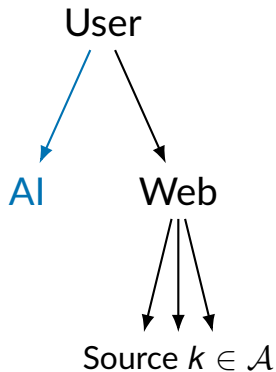
$$rs_k - F(1 - \eta) \geq 0,$$

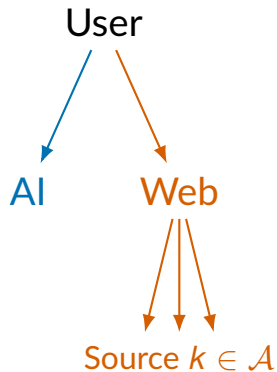
η being the cost reduction due to AI.



► Utility:

$$U_{iA} = \ln Q_A + \epsilon_{iA},$$



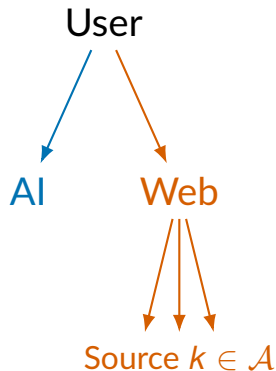


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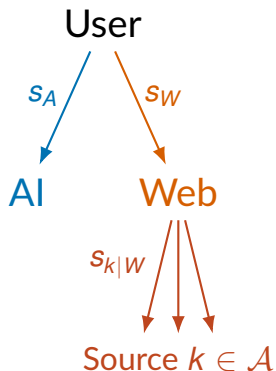
$$U_{ik} = \ln Q_k + \epsilon_{ik}.$$

$$\epsilon_i \sim GEV(\theta)$$

- AI aggregates the information available on the web, with efficiency ϕ

$$Q_A = \phi \left(\sum_{k \in \mathcal{A}} Q_k^{1/\theta} \right)^\theta.$$

Shares derivation



► The shares of users that go to:

► AI

$$s_A = \frac{\phi}{1 + \phi}$$

► Web

$$s_W = \frac{1}{1 + \phi}$$

► Source k

$$s_{k|W} = \frac{Q_k^{1/\theta}}{\sum_{j \in \mathcal{A}} Q_j^{1/\theta}}$$

Derive Q_0, m_A

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Supply of information:

- ▶ **increases** $\uparrow$ because AI decreases the cost of provision (η).
- ▶ **decreases** $\downarrow$ because AI steals the consumers, thus there is less incentive to produce (ϕ).

Assumption

Web/AI pieces

Derive $\tilde{Q}$

- ▶ We care about the quality that is consumed by the users:

$$\tilde{Q} \equiv s_A Q_A + \sum_{j \in A} s_j Q_j$$

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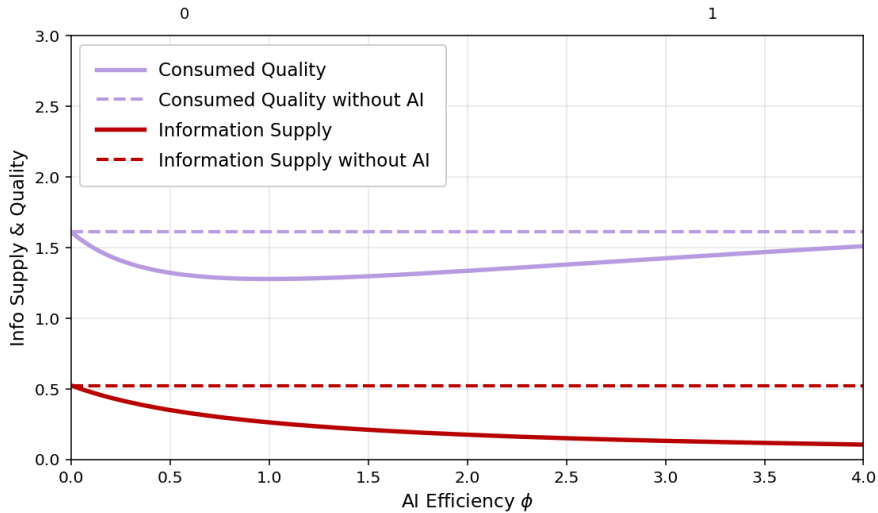
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Fixed Cost Reduction η

$\eta = 0.00$



Limitations & Solutions

- ▶ **AI's Impact on Creators**
- ▶ **Distributional Dependence**
- ▶ **Quality of AI**
- ▶ **Redirection and Google AIO**

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- ▶ **Problem:** Only $F(1 - \eta)$ not realistic

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- ▶ **Problems:**
 1. AI provides a redirection link to its sources.
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- ▶ **Solution:** Include a exogenous rates that transfer the users from AI to Web and vice versa.

GESIS Panel.dbd Digital Behavioral Data (Waiting for approval)

- ▶ 23 Feb 2023 to 20 Nov 2023.
- ▶ Recruitment from Meta ads ($n = 1,768$) and ALLBUS 2023 ($n = 800$).
- ▶ Data on visited websites with URLs, timestamps, and HTML page content.
- ▶ Linked to a survey about AI, Internet, and Socio-economic details.

Conclusion

- ▶ I want to investigate *how does Generative AI (Gen AI) affect online content supply and quality*
- ▶ For now, I have built a simple model, that predicts:
 - ▶ AI **increases** $\uparrow$ the **supply of information** because it decreases cost of production.
 - ▶ AI **decreases** $\downarrow$ the **supply of information** because it steals business.
 - ▶ If AI is good enough it **increases** the **quality of information**.
- ▶ In the future, confirm the predictions in the data (?).

Thank you!

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Appendix

[Parameters](#)[User shares](#)[Creator entry](#)[Web/AI pieces](#)[Quality derivation](#)

Click a button to jump to the relevant appendix block.

Appendix: parameters

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Symbol	Meaning	Notes
m	Mass of potential creators	ex ante
Q_k	Quality of creator k	$Q_k \sim \text{Pareto}(\gamma)$
γ	Pareto tail index	heterogeneity in quality
θ	Taste-shock / web-choice parameter	enters $Q_k^{1/\theta}$
F	Fixed operating cost	with AI: $(1 - \eta)F$
r	Revenue per unit of traffic	creator payoff scale
η	AI cost reduction	$\eta \in [0, 1)$
ϕ	AI efficiency / quality shifter	traffic-stealing force

Appendix: objects

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Symbol	Meaning	Notes
Q_0	Entry cutoff quality	active iff $Q_k \geq Q_0$
m_A	Share of active creators	$m_A = \Pr(Q \geq Q_0) = Q_0^{-\gamma}$
S	Web quality mass	$S \equiv \sum_{j \in A} Q_j^{1/\theta}$
Q_A	AI quality index	$Q_A = \phi S^\theta$
$\tilde{Q}$	Average consumed quality	$\tilde{Q} = s_A Q_A + \sum_{j \in A} s_j Q_j$
$s_{k W}$	Share on web	conditional on using the web
s_W, s_A	Web / AI shares	top-level channel shares
s_k	Creator k total share	$s_k = s_W s_{k W}$

Assumption $\gamma - 1/\theta - 1 > 0$

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For $Q \sim \text{Pareto}(\gamma)$ and conditioning on entry $Q \geq Q_0$,

$$f(q \mid q \geq Q_0) = \gamma Q_0^\gamma q^{-\gamma-1}, \quad q \geq Q_0.$$

Then

$$\mathbb{E}[Q^{1/\theta} \mid Q \geq Q_0] = \int_{Q_0}^{\infty} q^{1/\theta} \gamma Q_0^\gamma q^{-\gamma-1} dq = \frac{\gamma}{\gamma - 1/\theta} Q_0^{1/\theta},$$

which requires

$$\gamma - \frac{1}{\theta} > 0.$$

Also,

$$\mathbb{E}[Q^{1/\theta+1} \mid Q \geq Q_0] = \int_{Q_0}^{\infty} q^{1/\theta+1} \gamma Q_0^\gamma q^{-\gamma-1} dq.$$

Assumption $\gamma - 1/\theta - 1 > 0$ II

[▶ Previous](#)[▶ Back](#)

Continuing,

$$\mathbb{E}\left[Q^{1/\theta+1} \mid Q \geq Q_0\right] = \frac{\gamma}{\gamma - 1/\theta - 1} Q_0^{1/\theta+1},$$

which requires

$$\boxed{\gamma - \frac{1}{\theta} - 1 > 0.}$$

Hence

$$\frac{\mathbb{E}[Q^{1/\theta+1} \mid Q \geq Q_0]}{\mathbb{E}[Q^{1/\theta} \mid Q \geq Q_0]} = \frac{\gamma - 1/\theta}{\gamma - 1/\theta - 1} Q_0$$

is finite only under this assumption.

This is the condition needed for the web-quality term in $\tilde{Q}$ to be well defined.

User side: shares I

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Define

$$S \equiv \sum_{j \in A} Q_j^{1/\theta}.$$

Conditional on using the web,

$$s_{k|W} = \frac{e^{\ln Q_k / \theta}}{\sum_{j \in A} e^{\ln Q_j / \theta}} = \frac{Q_k^{1/\theta}}{\sum_{j \in A} Q_j^{1/\theta}} = \frac{Q_k^{1/\theta}}{S}.$$

Assume AI aggregates web content as

$$Q_A = \phi S^\theta.$$

The top-level web alternative has inclusive value S^θ .

User side: shares II

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Using the top-level values S^θ for the web and $Q_A = \phi S^\theta$ for AI,

$$s_W = \frac{S^\theta}{S^\theta + Q_A} = \frac{S^\theta}{S^\theta + \phi S^\theta} = \frac{1}{1 + \phi},$$

$$s_A = \frac{Q_A}{S^\theta + Q_A} = \frac{\phi S^\theta}{S^\theta + \phi S^\theta} = \frac{\phi}{1 + \phi}.$$

Therefore the unconditional traffic share of creator k is

$$s_k = s_W s_{k|W} = \frac{1}{1 + \phi} \frac{Q_k^{1/\theta}}{S}.$$

Creator side: Q_0 and m_A I

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Creators enter iff

$$\pi_k = rs_k - F(1 - \eta) \geq 0.$$

At the cutoff Q_0 ,

$$rs(Q_0) = F(1 - \eta), \quad r \frac{1}{1 + \phi} \frac{Q_0^{1/\theta}}{S} = F(1 - \eta).$$

Since $Q \sim \text{Pareto}(\gamma)$,

$$m_A \equiv \Pr(Q \geq Q_0) = Q_0^{-\gamma}, \quad mm_A = mQ_0^{-\gamma}$$

is the mass of active creators. Then

$$S = mm_A \mathbb{E} \left[Q^{1/\theta} \mid Q \geq Q_0 \right] = mQ_0^{-\gamma} \frac{\gamma}{\gamma - 1/\theta} Q_0^{1/\theta}.$$

Creator side: Q_0 and m_A II

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Using the previous expression for S ,

$$S = m \frac{\gamma}{\gamma - 1/\theta} Q_0^{1/\theta - \gamma}.$$

Substitute into the cutoff condition:

$$\begin{aligned} r \frac{1}{1 + \phi} \frac{Q_0^{1/\theta}}{m \frac{\gamma}{\gamma - 1/\theta} Q_0^{1/\theta - \gamma}} &= F(1 - \eta) \\ \Rightarrow Q_0^\gamma &= \frac{Fm\gamma}{r(\gamma - 1/\theta)} (1 - \eta)(1 + \phi). \end{aligned}$$

Therefore

$$Q_0 = \left[\frac{Fm\gamma}{r(\gamma - 1/\theta)} (1 - \eta)(1 + \phi) \right]^{1/\gamma}$$

and

$$m_A = Q_0^{-\gamma} = \left[\frac{Fm\gamma}{r(\gamma - 1/\theta)} (1 - \eta)(1 + \phi) \right]^{-1}.$$

Web quality and AI quality I

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Web quality consumed conditional on using the web is

$$\sum_{j \in A} s_{j|W} Q_j = \frac{\sum_{j \in A} Q_j^{1/\theta+1}}{\sum_{j \in A} Q_j^{1/\theta}} = \frac{\mathbb{E}[Q^{1/\theta+1} \mid Q \geq Q_0]}{\mathbb{E}[Q^{1/\theta} \mid Q \geq Q_0]} = \frac{\gamma - 1/\theta}{\gamma - 1/\theta - 1} Q_0.$$

Hence the web contribution to average consumed quality is

$$s_W \sum_{j \in A} s_{j|W} Q_j = \frac{Q_0}{1 + \phi} \frac{\gamma - 1/\theta}{\gamma - 1/\theta - 1}.$$

Web quality and AI quality II

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Use the cutoff condition to rewrite S :

$$r \frac{1}{1+\phi} \frac{Q_0^{1/\theta}}{S} = F(1-\eta) \quad \Rightarrow \quad S = \frac{r}{F(1-\eta)(1+\phi)} Q_0^{1/\theta}.$$

Since

$$Q_A = \phi S^\theta,$$

we get

$$Q_A = \phi \left(\frac{r}{F(1-\eta)(1+\phi)} \right)^\theta Q_0.$$

Therefore the AI contribution is

$$s_A Q_A = \frac{\phi}{1+\phi} \phi \left(\frac{r}{F(1-\eta)(1+\phi)} \right)^\theta Q_0 = \frac{Q_0}{1+\phi} \phi^2 \left(\frac{r}{F(1-\eta)(1+\phi)} \right)^\theta.$$

Derivation of $\tilde{Q}$ I

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Start from

$$\tilde{Q} = s_A Q_A + \sum_{j \in A} s_j Q_j.$$

Use

$$\sum_{j \in A} s_j Q_j = s_W \sum_{j \in A} s_{j|W} Q_j = \frac{Q_0}{1 + \phi} \frac{\gamma - 1/\theta}{\gamma - 1/\theta - 1},$$

and

$$s_A Q_A = \frac{Q_0}{1 + \phi} \phi^2 \left(\frac{r}{F(1 - \eta)(1 + \phi)} \right)^\theta.$$

Derivation of $\tilde{Q}$ II

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Combining terms,

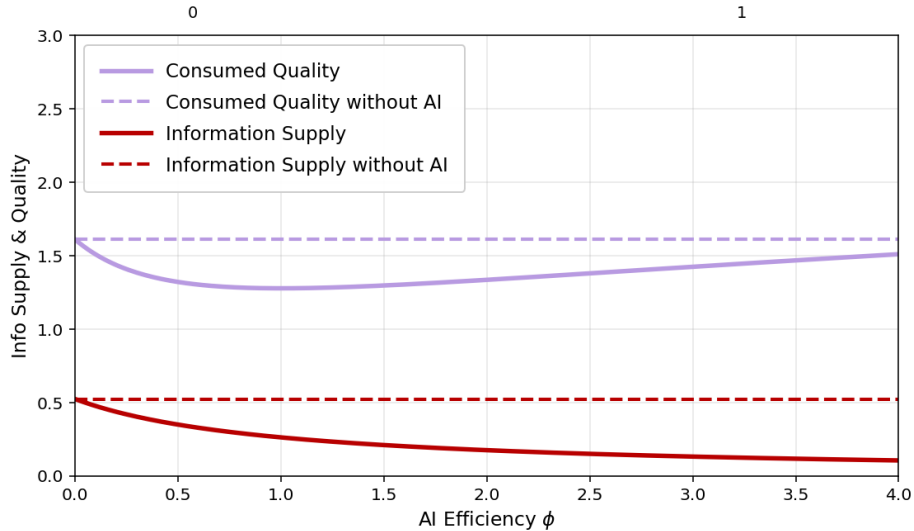
$$\tilde{Q} = \frac{Q_0}{1+\phi} \left[\frac{\gamma - 1/\theta}{\gamma - 1/\theta - 1} + \phi^2 \left(\frac{r}{F(1-\eta)(1+\phi)} \right)^\theta \right].$$

With

$$Q_0 = \left[\frac{Fm\gamma}{r(\gamma - 1/\theta)} (1-\eta)(1+\phi) \right]^{1/\gamma}, \quad m_A = Q_0^{-\gamma}.$$

Fixed Cost Reduction η

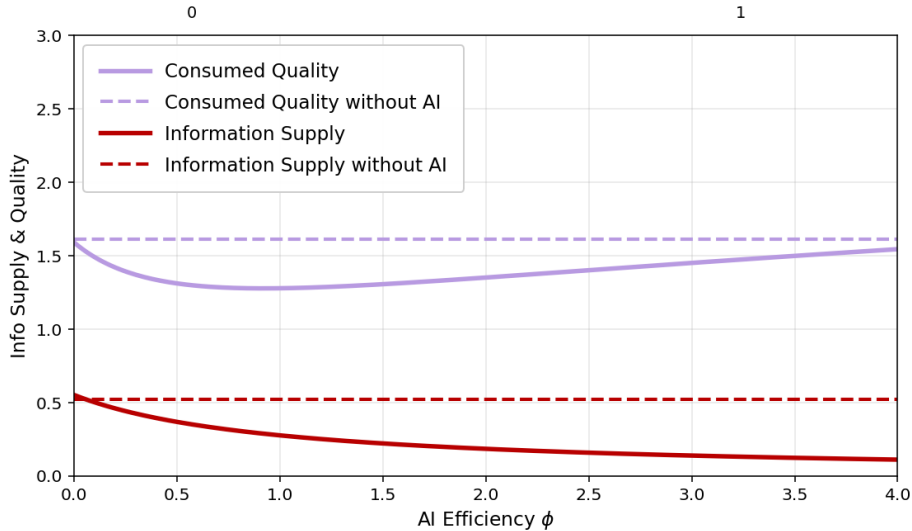
$\eta = 0.00$



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Fixed Cost Reduction η

$\eta = 0.05$



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Fixed Cost Reduction η

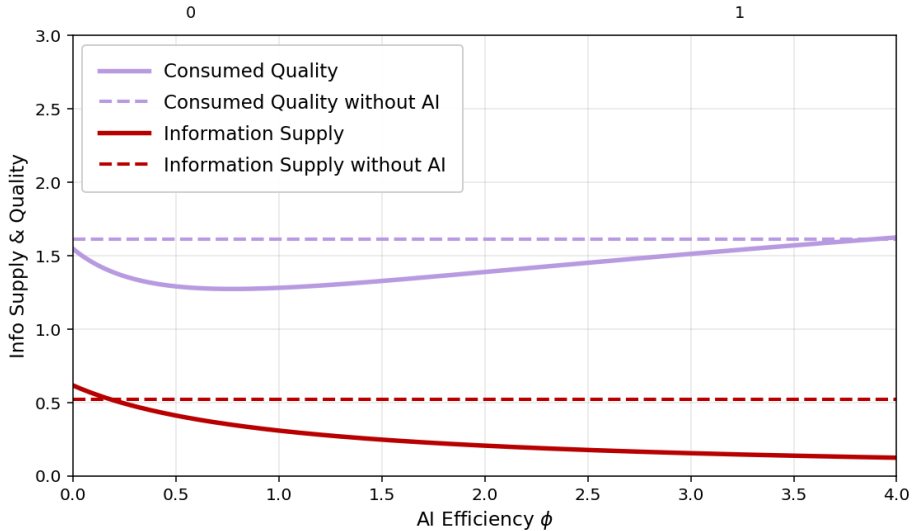
$\eta = 0.10$



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Fixed Cost Reduction η

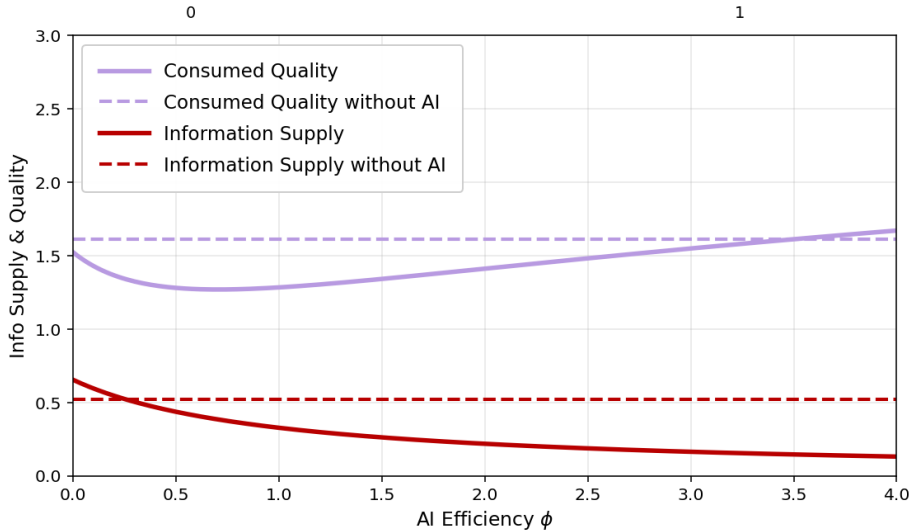
$\eta = 0.15$



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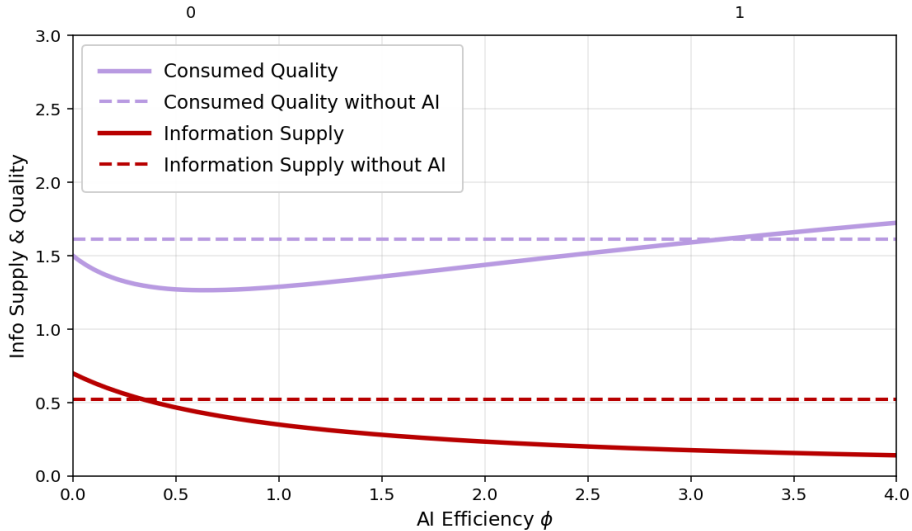
Fixed Cost Reduction η

$\eta = 0.20$



Fixed Cost Reduction η

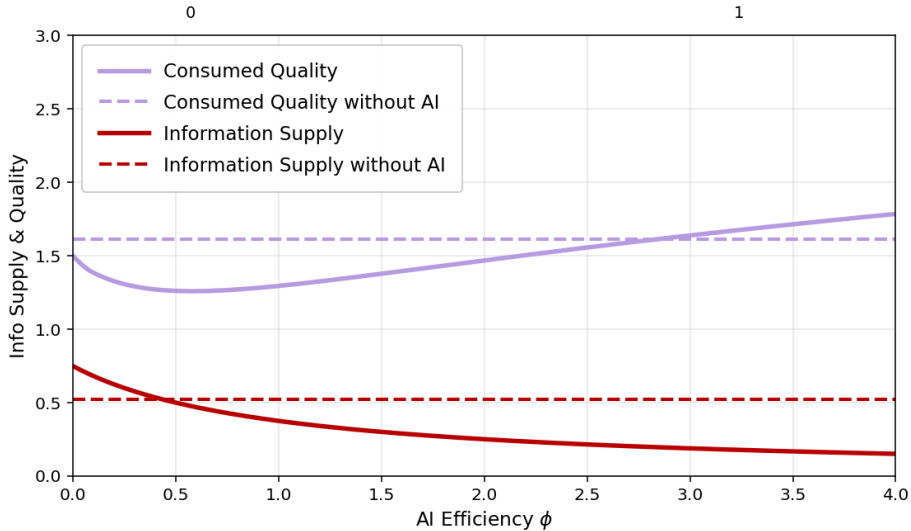
$\eta = 0.25$



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Fixed Cost Reduction η

$\eta = 0.30$



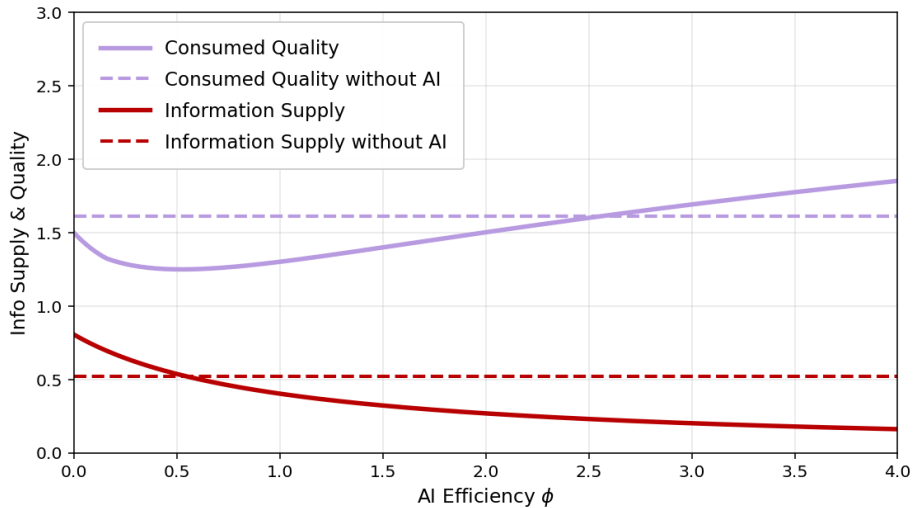
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Fixed Cost Reduction η

$\eta = 0.35$

0

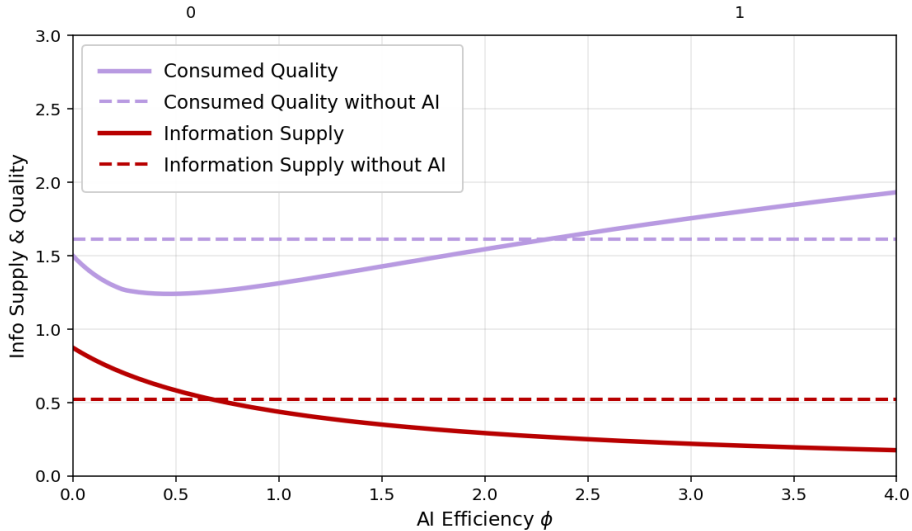
1



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Fixed Cost Reduction η

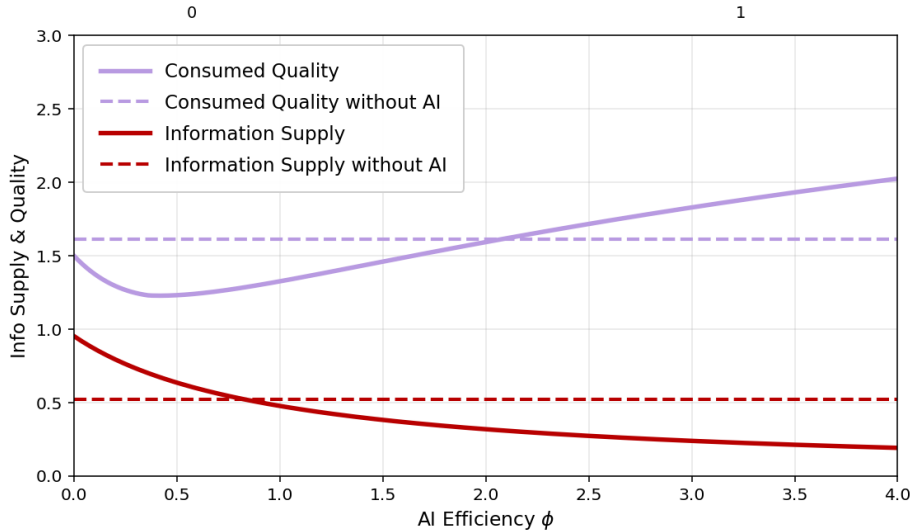
$\eta = 0.40$



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Fixed Cost Reduction η

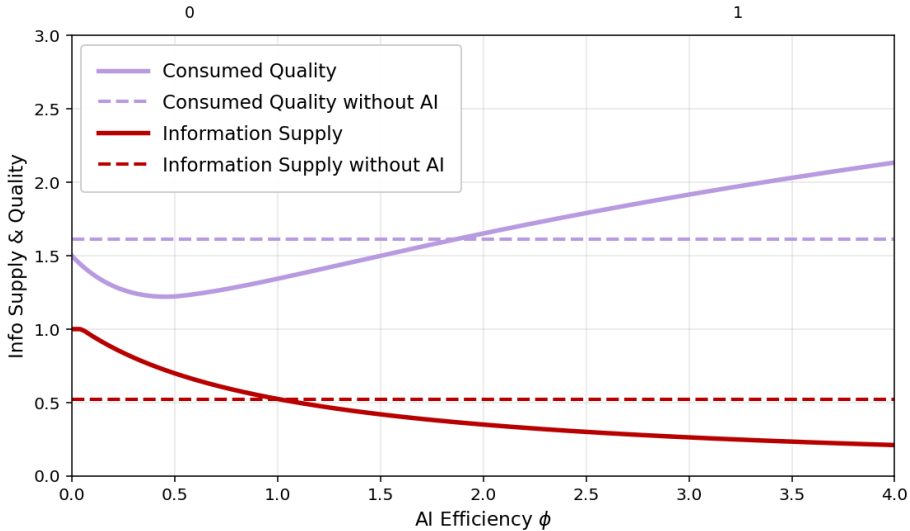
$\eta = 0.45$



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Fixed Cost Reduction η

$\eta = 0.50$



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Fixed Cost Reduction η

$\eta = 0.55$



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Fixed Cost Reduction η

$\eta = 0.60$



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Fixed Cost Reduction η

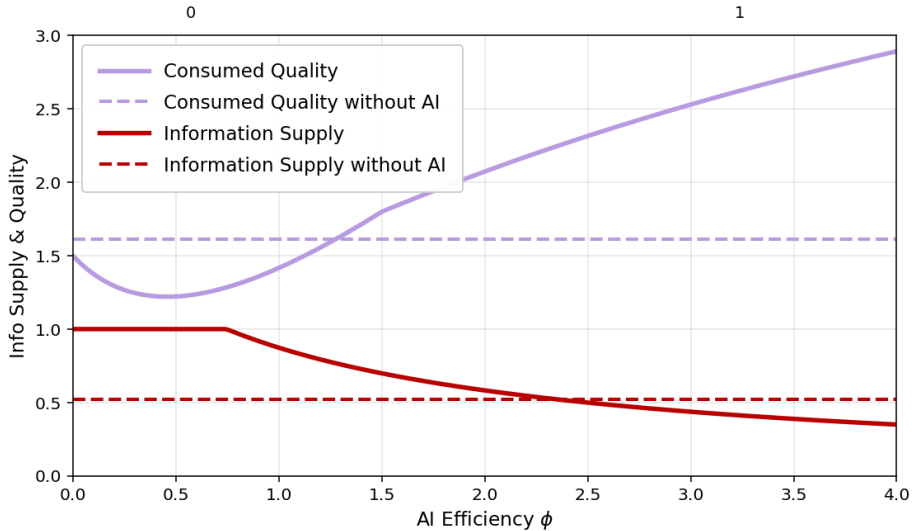
$\eta = 0.65$



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Fixed Cost Reduction η

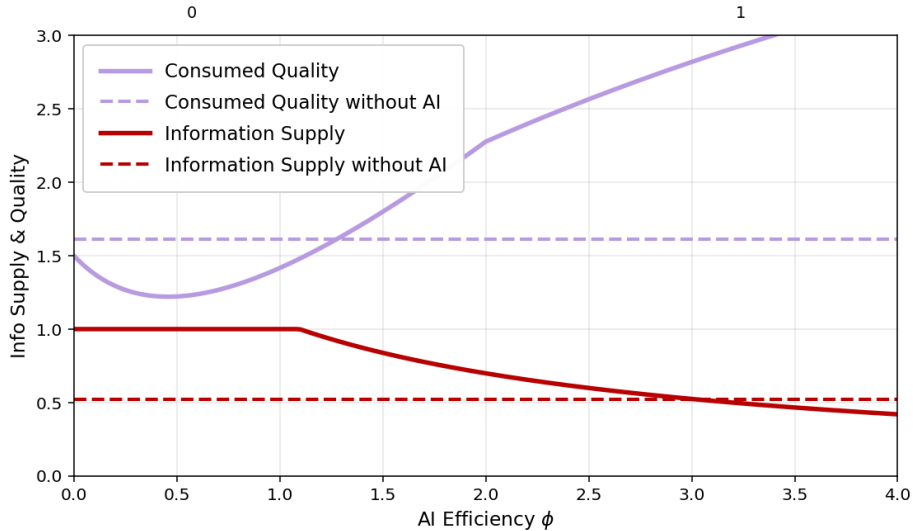
$\eta = 0.70$



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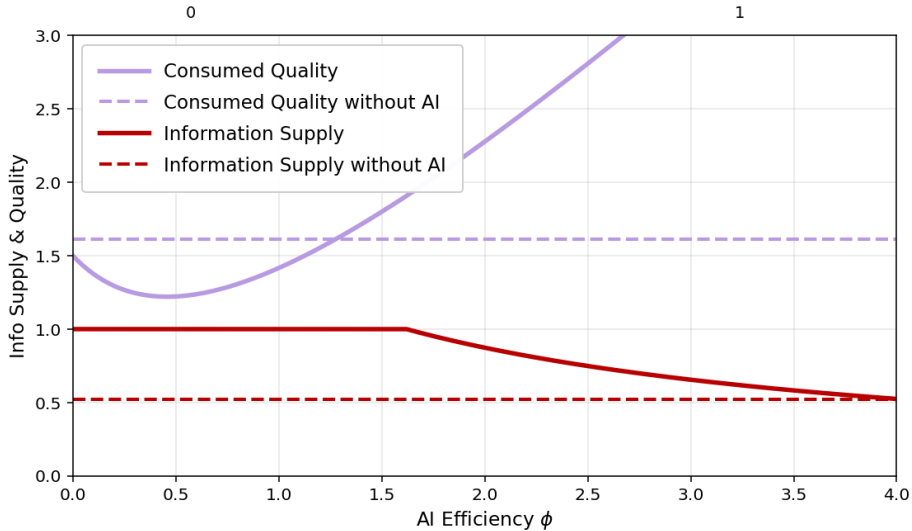
Fixed Cost Reduction η

$\eta = 0.75$



Fixed Cost Reduction η

$\eta = 0.80$



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Fixed Cost Reduction η

$\eta = 0.85$

